

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

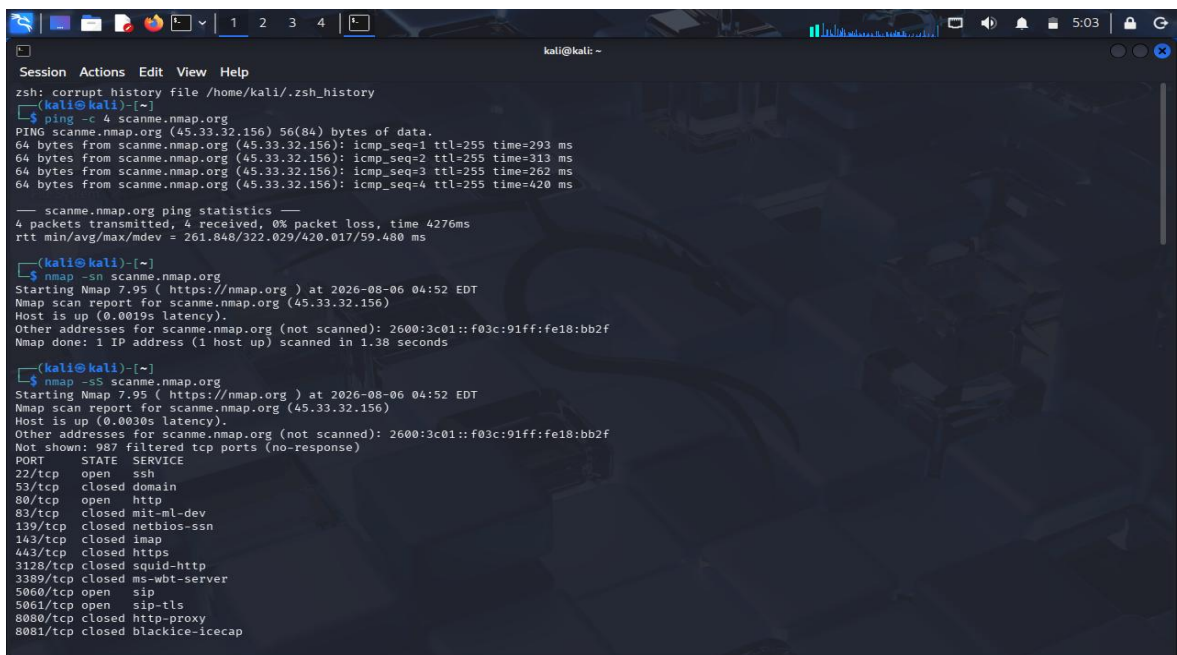
CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ ping -c 4 scanme.nmap.org
PING scanme.nmap.org (45.33.32.156): 56(84) bytes of data:
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=255 time=293 ms
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=2 ttl=255 time=313 ms
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=3 ttl=255 time=262 ms
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=4 ttl=255 time=420 ms

--- scanme.nmap.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4276ms
rtt min/avg/max/mdev = 261.848/322.029/420.017/59.480 ms

(kali@kali)~$ nmap -sn scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 04:52 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0019s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 1.38 seconds

(kali@kali)~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 04:52 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0030s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
83/tcp    closed mit-ml-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
```

2. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and

Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.

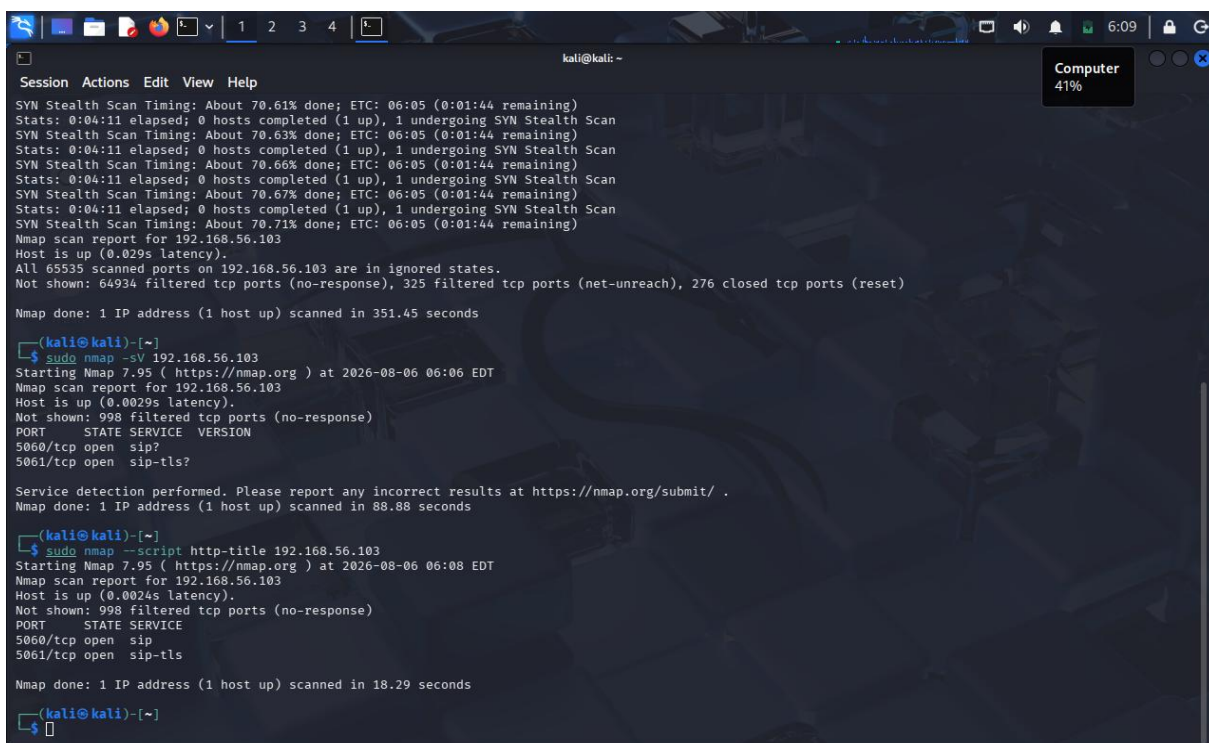
```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 12.11 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sV 192.168.56.101  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:51 EDT  
Nmap scan report for 192.168.56.101  
Host is up (0.0012s latency).  
All 1000 scanned ports on 192.168.56.101 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.41 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sC 192.168.56.101  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:52 EDT  
Nmap scan report for 192.168.56.101  
Host is up (0.00046s latency).  
All 1000 scanned ports on 192.168.56.101 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 16.35 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.56.101  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:52 EDT  
Nmap scan report for 192.168.56.101  
Host is up (0.0015s latency).  
All 1000 scanned ports on 192.168.56.101 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox  
Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%)  
, HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT ADDRESS  
1 1.05 ms 192.168.56.101  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 28.04 seconds
```

3. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 4.74 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sV 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:17 EDT  
Nmap scan report for 192.168.56.102  
Host is up (0.0047s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 72.46 seconds  
  
(kali@kali)-[~]  
$ sudo nmap --script rdp-enum-encryption 192.168.56.102  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:19 EDT  
Nmap scan report for 192.168.56.102  
Host is up (0.0045s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 4.32 seconds  
  
(kali@kali)-[~]  
$ sudo nmap --script smb-enum-shares 192.168.56.102  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:19 EDT  
Nmap scan report for 192.168.56.102  
Host is up (0.0078s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 4.37 seconds  
  
(kali@kali)-[~]  
$
```

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating

system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure



```
Session Actions Edit View Help
SYN Stealth Scan Timing: About 70.61% done; ETC: 06:05 (0:01:44 remaining)
Stats: 0:04:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.63% done; ETC: 06:05 (0:01:44 remaining)
Stats: 0:04:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.66% done; ETC: 06:05 (0:01:44 remaining)
Stats: 0:04:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.67% done; ETC: 06:05 (0:01:44 remaining)
Stats: 0:04:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.71% done; ETC: 06:05 (0:01:44 remaining)
Nmap scan report for 192.168.56.103
Host is up (0.029s latency).
All 65535 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 64934 filtered tcp ports (no-response), 325 filtered tcp ports (net-unreach), 276 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 351.45 seconds

(kali@kali)-[~]
└─$ sudo nmap -sV 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:06 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0029s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

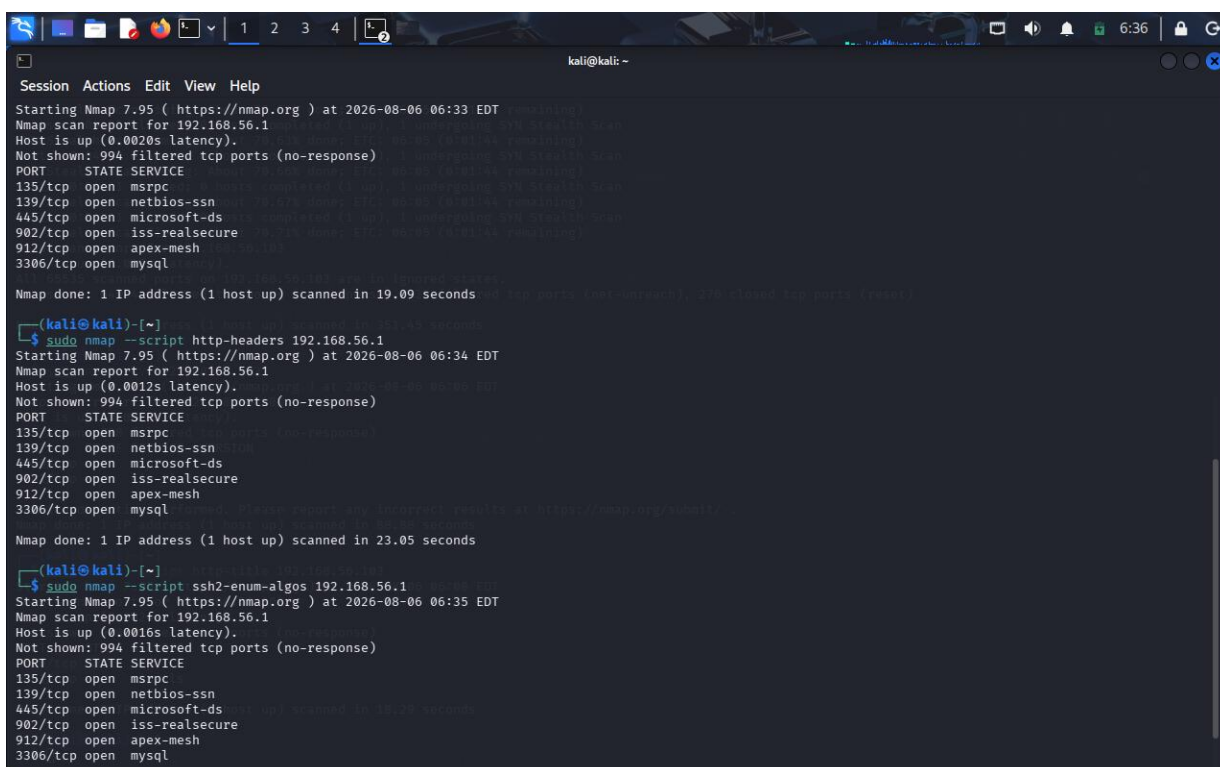
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 88.88 seconds

(kali@kali)-[~]
└─$ sudo nmap --script http-title 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:08 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0024s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 18.29 seconds

(kali@kali)-[~]
└─$
```

5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.



```
Session Actions Edit View Help
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:33 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0020s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsure
912/tcp   open  apex-mesh
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 19.09 seconds

(kali@kali)-[~]
└─$ sudo nmap --script http-headers 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:34 EDT
Nmap scan report for 192.168.56.1
Host is up (0.002s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsure
912/tcp   open  apex-mesh
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 23.05 seconds

(kali@kali)-[~]
└─$ sudo nmap --script ssh2-enum-algos 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 06:35 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
Not shown: 994 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsure
912/tcp   open  apex-mesh
3306/tcp  open  mysql
```

Code of Conduct:

I k. Venkata Karthik 192372092 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Venkata Karthik

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	